Guidelines for Improving Arabic Pronunciation in AI Speech Systems

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Objective: Improving pronunciation accuracy and naturalness in Arabic AI speech systems through enhanced linguistic preprocessing.

1. Introduction

Recent advances in neural speech synthesis have significantly improved the quality of AI-generated speech in many languages. However, Arabic speech produced by many AI systems still falls short of the level of natural pronunciation expected by native speakers.

In many cases, these shortcomings are not caused by limitations in the acoustic speech generation models themselves, but rather by insufficient linguistic preprocessing of the input text before it is passed to the speech synthesis system.

Arabic is a morphologically rich language with complex phonological rules, contextual vowelization, and syntactic dependencies that directly affect pronunciation. When these linguistic aspects are not properly handled, the resulting speech may sound unnatural or even convey incorrect meanings.

This document proposes a structured **Pre-Pronunciation Processing Layer** designed to improve Arabic speech synthesis by addressing common linguistic and phonological

issues before speech generation occurs.
The proposed approach focuses on improving pronunciation accuracy through:

- text normalization
- contextual morphological analysis
- automatic diacritization
- rule-based pronunciation adjustments

By implementing this preprocessing layer, AI speech systems can significantly improve pronunciation accuracy and naturalness for Arabic speakers.

2. Proposed Architecture

The proposed architecture introduces a linguistic preprocessing stage before speech synthesis.
Processing pipeline:
Raw Text
↓
Text Normalization
↓
Context-Aware Morphological Analysis
↓
Arabic Diacritization
↓
Pronunciation Rules Processing
↓
Speech Synthesis
This architecture ensures that linguistic ambiguities are resolved before phonetic generation begins.

3. Rule 1 — Number Normalization and Agreement

Numbers should never be pronounced directly as digits.

Instead, they must be converted into their lexical Arabic forms before speech synthesis.
Example:
DigitArabic Form1 واحد2اثنان3ثلاثة4أربعة5خمسة10عشرة11أحد عشر12اثنا عشر13ثلاثة عشر14أربعة عشر
Arabic numbers also require grammatical agreement with the counted noun.
Example:
3 students → ثلاثة طلاب
3 schools → ثلاث مدارس
Numbers from **3–10 follow a reverse gender agreement rule** with the counted noun, which must be handled during preprocessing.
Future implementations should also support:

- compound numbers (21–99)
- ordinal numbers (الأول، الثاني)
- fractional numbers (نصف، ثلث، ربع)

4. Rule 2 — Context-Aware Morphological Analysis

Arabic contains many homographs — words that share identical spelling but differ in pronunciation depending on context.
Example:
ملك
Possible interpretations:

- (king) مَلِك
- (angel) مَلَك
- (ownership) مُلْك

Resolving these ambiguities requires syntactic and semantic analysis using techniques commonly applied in Natural

Language Processing.
Tools such as:

- part-of-speech tagging
- named entity recognition
- contextual language models

can significantly improve disambiguation accuracy.

5. Rule 3 — Arabic Diacritization

After contextual analysis, the system must automatically add diacritics to the text before generating speech.
This process is known as **Arabic diacritization**.
Example:
زار الملك المدينة
becomes
زارَ المَلِكُ المدينةَ
Diacritization provides the phonological information required for accurate pronunciation.

6. Rule 4 — Sun Letters and Moon Letters

The Arabic definite article ال behaves differently depending on the following consonant.

Sun Letters

The consonant **lām (ل)** is assimilated to the following sun letter, and the consonant is geminated.
Example:
ash-shams → الشمس
adh-dhakāʾ → الذكاء

Moon Letters

The **lām** sound remains pronounced.
Example:

al-qamar → القمر
al-kitāb → الكتاب

7. Rule 5 — Shadda (Gemination)

The diacritic **shadda (ّ)** indicates consonant doubling.
Example:
mad + d → مدّ
ṣad + daq → صدّق
Failure to correctly realize gemination results in unnatural rhythm and pronunciation.

8. Rule 6 — Taa Marbuta (ة)

The pronunciation of **taa marbuta (ة)** depends on syntactic context.

Pause position

Pronounced as **h**
Example:
madrasa(h) → مدرسة

Construct state (Idāfah)

Pronounced as **t**
Example:
madrasat al-ʿulūm → مدرسة العلوم

9. Rule 7 — Alif Maqsura (ى)

The letter ى represents a long vowel **ā**.
Examples:
hudā → هدى
muṣṭafā → مصطفى
ʿalā → على
It must never be pronounced as a consonantal **y**.

10. Rule 8 — Nunation (Tanween)

Nunation markers (ـٌ ـٍ ـً) influence both grammatical meaning and phonetic rhythm.

Examples:

كتابٌ

كتابًا

كتابٍ

The system should correctly realize nunation in connected speech while allowing optional suppression in pause position.

11. Rule 9 — Hamza Pronunciation

The glottal stop **hamza (ء)** plays a critical role in Arabic phonology.

Examples:

سؤال

قراءة

مؤمن

The system must distinguish between:

- **hamzat al-qaṭʿ**
- **hamzat al-waṣl**

Improper handling of hamza frequently leads to unnatural pronunciation.

12. Rule 10 — Long Vowel Duration

Arabic contains three primary long vowels:

- ā
- ī
- ū

Example:

قال

قيل

يقول

Incorrect vowel duration significantly affects perceived naturalness.

13. Rule 11 — Stress Placement

Arabic stress patterns follow predictable phonological rules. Incorrect stress placement produces speech that sounds foreign to native speakers.

Stress modeling should therefore be incorporated within pronunciation rules.

14. Preservation of Modern Standard Arabic Phonemes

To maintain intelligibility across the Arab world, pronunciation should follow **Modern Standard Arabic (MSA)** phonology.

Examples:

/q/ → ق

/ʕ/ → ع

/ħ/ → ح

Dialect substitutions should be avoided unless the system explicitly targets a specific regional dialect.

15. Evaluation Methodology

System performance should be evaluated using both objective and subjective methods.

Suggested metrics include:

- phoneme accuracy rate
- pronunciation error frequency
- Mean Opinion Score (MOS) from native Arabic listeners

Testing should include speakers from different Arabic regions to ensure broad intelligibility.

16. Observed Pronunciation Errors

in Current AI Systems

Direct interaction with multiple AI speech systems reveals several recurring pronunciation errors.
These errors typically result from missing or incomplete linguistic preprocessing.
Examples include:

Homograph misinterpretation

ملك
Incorrect pronunciation:
مَلَك
Correct pronunciation:
مَلِك
Example sentence:
زار الملك المدينة
Incorrect interpretation may change the meaning from **“the king visited the city”** to **“the angel visited the city.”**

Incorrect diacritization

النطاق → incorrectly pronounced as → النطق
Correct form:
النُّطق

Passive vs active confusion

طبقت
Possible forms:
طَبَّقت
طُبِّقت
Incorrect detection can change the intended meaning.

Taa marbuta errors

مدرسة العلوم
Correct pronunciation:

madrasat al-ʿulūm

17. Conclusion

Arabic pronunciation errors in AI speech systems often originate from incomplete linguistic preprocessing rather than limitations in speech synthesis models.

By introducing a structured **Pre-Pronunciation Processing Layer**, AI speech systems can significantly improve pronunciation accuracy, intelligibility, and naturalness for Arabic speakers.

Future work should focus on integrating these linguistic rules with modern language models and speech synthesis systems to create more natural and reliable Arabic AI speech.